



National University

of computer and emerging sciences

Student Name: _____

Roll No: _____

Program: BS(CS) -3c

Quiz : 1 (Easy one)

Instructor: Sir Samin Ahmed

Course: Data Structure

Date:

Time allowed: 20 Mins

Question 1: Write the output in front of each **cout** statement.

```
#include <bits/stdc++.h>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
int arr[10] = {0};
```

```
int *p = arr;
```

```
for (int i = 0; i < 10; i++)
```

```
{
```

```
cout << *p << " ";
```

```
p++;
```

```
}
```

```
p = p - 10;
```

```
p = p + 5;
```

```
*p = 100;
```

```
p = p - 5;
```

```
*p = 1;
```

```
int index = ((p - arr) + 10) % 10;
```

```
p = arr + index;
```

```
*p = 101;
```

```
p=arr+10-2;
```

```
*p=-1;
```

```
cout << "\nArray after modifications:\n";
```

```
for (int i = 0; i < 10; i++) cout << arr[i] << " ";
```

```
}
```

Hint: Don't let the pointers run away

Question 2

You are given an array of N elements in the below function with value k rotate the array to right k steps using only pointers no index is allowed.

Example :

Input: arr = [1, 2, 3, 4, 5], N = 5, K = 2

Output: arr = [4, 5, 1, 2, 3]

```
void rotateright(int arr[],int N,int K){
```

Hint: Use your brain.

}